

torch_vn_algebra: batched random operators, states, channels and constrained optimisation in finite-dimensional Type I von Neumann algebras

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Abstract

`torch_vn_algebra` is a PyTorch library for numerical work in algebras $\mathcal{M} = \bigoplus_{c=1}^C M_{k_c}(\mathbb{C})$, the finite-dimensional Type I von Neumann algebras that describe quantum systems with superselection rules. Operators, states and channels are stored as batched, block-diagonal tensors, so that thousands of random samples, or thousands of independent starts of an optimisation, are processed at once on a CPU or a GPU. The library provides random operators with prescribed spectra and Haar-distributed eigenvectors, the three trace functionals that are natural on $\mathcal{M}$, states and Lüders conditioning, completely positive maps within and between sectors, plain and charge-conserving (fused) tensor products with partial traces, spin chains with a conserved magnetisation, Hamiltonian and Lindblad dynamics, differentiable parametrisations with a prescribed Michelson contrast, and safeguards that estimate run time and memory before expensive steps. We describe the design, show how the library is used on short complete examples, and validate it against exact results from random matrix theory, quantum information and integrable spin chains (free-fermion spectra and dynamics, Heisenberg ring energies, Peschel’s entanglement formula, the GOE–Poisson crossover of disordered chains). Applied to the trace inequalities of our earlier work, constrained optimisation replaces Monte Carlo sampling: it recovers the sharp bounds that sampling misses in all but the smallest dimensions.

1 Introduction

Block-diagonal operators appear whenever a symmetry or a superselection rule splits the Hilbert space into sectors: charge or particle-number sectors, parity, total spin, or the classical part of a hybrid classical–quantum system. Mathematically, the observables form a finite-dimensional Type I von Neumann algebra $\mathcal{M} = \bigoplus_c M_{k_c}(\mathbb{C})$, whose centre is spanned by the sector projections, and the choice of trace on $\mathcal{M}$ is no longer unique. The questions we had in mind when writing this library come from operator theory: which inequalities between traces of products characterise the trace or the centre of $\mathcal{M}$ [2, 3, 4, 5], and how far do they fail in terms of a spectral quantity such as the Michelson contrast [6, 7]?

General-purpose quantum software (QuTiP [8], Qiskit [9], the JAX-based `dynamics` [10]) is built around a single Hilbert space; sectors, several traces and large batches of random operators with prescribed spectra are not first-class objects there. The niche of `torch_vn_algebra` is this combination: the algebra $\mathcal{M}$ and its traces as the basic objects, batched random sampling and batched optimisation over them, and enough quantum information, many-body structure and dynamics (states, channels, tensor products that merge sectors of equal charge, spin chains, Lindblad evolution) to formulate physical questions. We do not aim to replace the packages above for single-system simulations.

This text supersedes the description of the first version of the library [1]; the list of corrections to that version is part of the repository.

Contributions. (i) A batched block-diagonal data model with three trace conventions and explicit conversions between them (Sections 2, 3). (ii) States, channels within and between sectors, tensor products and dynamics that respect the sector structure (Section 4). (iii) A validation suite against exact results (Section 5). (iv) Differentiable parametrisations with a hard Michelson-contrast constraint and batched multistart optimisation, applied to sharp bounds of trace inequalities (Section 6). (v) Charge-conserving tensor products and $U(1)$ -symmetric spin chains, checked against exact results (Section 7). (vi) Run-time and memory safeguards (Section 3).

2 Mathematical setting

Algebra and traces. Let $\mathcal{M} = \bigoplus_{c=1}^C M_{k_c}(\mathbb{C})$ act on $\mathcal{H} = \bigoplus_c \mathbb{C}^{k_c}$; $A \in \mathcal{M}$ is block diagonal, $A = \bigoplus_c A_c$. Every trace on $\mathcal{M}$ is $\text{tr}_w(A) = \sum_c w_c \text{Tr } A_c$ with weights $w_c > 0$. The library implements

$$\text{Tr}_{\text{blunt}} : w_c = 1, \quad \text{Tr}_{\text{norm}} : w_c = \frac{1}{k_c}, \quad \tau_{\text{vN}} : w_c = \frac{1}{Ck_c},$$

where τ_{vN} is the faithful normal tracial state that gives every sector the same weight. For $A \in \mathcal{M}$, $|A| = (A^*A)^{1/2}$; for $A \geq 0$ the Michelson contrast is $\Delta(A) = (\lambda_{\max} - \lambda_{\min})/(\lambda_{\max} + \lambda_{\min}) \in [0, 1]$, with the extreme eigenvalues taken over all sectors; $\Delta(A) = 0$ if and only if A is a multiple of the identity.

States. A state ω on $\mathcal{M}$ is represented by its density $\rho \geq 0$ with respect to Tr_{blunt} , $\omega(A) = \text{Tr}(\rho A)$, $\sum_c \text{Tr } \rho_c = 1$. The restriction of ω to the centre is the classical distribution $p_c = \text{Tr } \rho_c$ over sectors. The density of the same state with respect to tr_w is $\rho_c^{(w)} = \rho_c/w_c$; the library converts between the conventions explicitly, because statements such as “trace preserving” or “the dual map” depend on them as soon as the sectors change.

Maps between sectors. Let $\mathcal{M}_{\text{in}} = \bigoplus_c M_{k_c}$ and $\mathcal{M}_{\text{out}} = \bigoplus_d M_{m_d}$. Every normal completely positive map $\Phi : \mathcal{M}_{\text{in}} \rightarrow \mathcal{M}_{\text{out}}$ (in the Schrödinger picture) is a sum of completely positive maps between blocks,

$$\Phi(\rho)_d = \sum_c \sum_i K_i^{dc} \rho_c (K_i^{dc})^*, \quad K_i^{dc} : \mathbb{C}^{k_c} \rightarrow \mathbb{C}^{m_d}.$$

It preserves Tr_{blunt} iff $\sum_{d,i} (K_i^{dc})^* K_i^{dc} = 1_c$ for all c , and $T_{dc} = \text{Tr } \Phi_{dc}(\rho_c)/\text{Tr } \rho_c$ is the classical transition matrix between sectors. The dual with respect to tr_w on both sides is $\Phi^{*,w}(A)_c = \sum_{d,i} (w_d^{\text{out}}/w_c^{\text{in}})(K_i^{dc})^* A_d K_i^{dc}$; for sector-preserving maps on one algebra the weights cancel and all three traces give the same dual.

Tensor products and fusion of sectors. $(\bigoplus_c M_{k_c}) \otimes (\bigoplus_d M_{m_d}) = \bigoplus_{(c,d)} M_{k_c m_d}$ is again of Type I with sectors labelled by pairs. If the sectors carry conserved charges q_c, q'_d (particle numbers, magnetisation), the operators of the composite system that conserve only the *total* charge form the larger algebra

$$\mathcal{M}_1 \hat{\otimes} \mathcal{M}_2 = \bigoplus_q M_{n_q}, \quad n_q = \sum_{q_c + q'_d = q} k_c m_d,$$

in which all pairs with the same total charge are merged into one sector. It contains $\mathcal{M}_1 \otimes \mathcal{M}_2$ (block-diagonally in every merged sector), but also operators such as $S_i^+ S_j^-$ that move charge between the factors. For additive charges, the partial trace $(\text{Tr}_2 X)_c = \sum_d \text{Tr}_{m_d} X_{(c,d),(c,d)}$ still lands in $\mathcal{M}_1$: a block between pairs $(c,d) \neq (c',d)$ with equal total charge would need $c = c'$. In both cases the partial trace is a map between sectors in the sense above, and its dual with respect to Tr_{blunt} is the embedding $A \mapsto A \otimes 1$.

Spin chains. For L spins $1/2$ with conserved S_{tot}^z , iterating the fused product of single-site algebras $M_1 \oplus M_1$ (charges $0, 1$) gives $\mathcal{M} = \bigoplus_{N=0}^L M_{\binom{L}{N}}(\mathbb{C})$, N the number of up spins, which contains every S_{tot}^z -conserving Hamiltonian, e.g. the XXZ chain $H = \sum_{\langle ij \rangle} [\frac{J}{2}(S_i^+ S_j^- + S_i^- S_j^+) + J\Delta S_i^z S_j^z] + \sum_i h_i S_i^z$.

Dynamics. For $H = H^* \in \mathcal{M}$ and jump operators $L_j \in \mathcal{M}$ the Gorini–Kossakowski–Sudarshan–Lindblad equation $\dot{\rho} = -i[H, \rho] + \sum_j \gamma_j (L_j \rho L_j^* - \frac{1}{2}\{L_j^* L_j, \rho\})$ preserves every sector, so the sector probabilities p_c are conserved (superselection).

3 Design

Data layout. An operator is a tensor of shape $(B, C, k_{\text{max}}, k_{\text{max}})$: B samples, C sectors, block c in the top-left $k_c \times k_c$ corner and zeros elsewhere. Linear algebra is applied to the batch of active blocks of each sector, so one call to a batched LAPACK/cuSOLVER routine serves all samples. The padding costs memory when the sector sizes differ strongly (for a chain of 14 spins, 77 % of a padded operator is zeros); the library warns in that case, and `SpinChain` lets one work in single sectors $M_{\binom{L}{N}}$ without padding, which is what dense diagonalisation needs anyway.

Objects. `Operator` (with `DensityMatrix` as a subclass for states), `Channel` (sector preserving) and `InterSectorChannel` (between sectors or algebras) are thin wrappers around such tensors; Table 1 lists the modules. Operations on operators are lazy: they record how to compute the result and do so on first use; the result is cached and the recipe dropped, so intermediates of long expressions can be freed. Laziness defers and skips work but does not fuse operations; where this matters the library uses direct formulas, e.g. $\text{Tr}(\rho A) = \sum_{ij} \rho_{ij} A_{ji}$ in $O(k^2)$ instead of forming ρA .

module	content
<code>algebra</code>	<code>TypeIAlgebra</code> , <code>Operator</code> : random operators, unitary ensembles, traces, functional calculus
<code>states</code>	<code>DensityMatrix</code> , random states, Gibbs states, Lüders rule, entropies, fidelity
<code>channels</code>	<code>Channel</code> , <code>InterSectorChannel</code> , Choi matrices, standard and random channels
<code>composite</code>	plain and fused (charge-conserving) tensor products, <code>kron</code> , partial traces
<code>chains</code>	<code>SpinChain</code> : $U(1)$ sectors, XXZ Hamiltonians with batched disorder, reduced states
<code>dynamics</code>	exact Schrödinger/von Neumann evolution, RK4, Lindblad equation, $e^{t\mathcal{L}}$
<code>optimize</code>	parametrisations with prescribed contrast, batched multistart <code>extremize</code>
<code>cost</code>	run-time and memory estimates, warnings and errors

Table 1: Modules of `torch_vn_algebra`.

Random unitaries. Haar unitaries on $U(n)$ and $O(n)$ are obtained from the QR decomposition of a Ginibre matrix with i.i.d. circular Gaussian entries after fixing the phases of $\text{diag } R$ [11]; $SU(n)$ and $SO(n)$ by multiplying with $\det^{-1/n}$ or by a sign flip of one column; the circular orthogonal and symplectic ensembles as $W^T W$ and $W^R W$ with W Haar on $U(n)$ [12]. Section 5 checks all of them.

Differentiation. All operations are composed of differentiable PyTorch primitives, so gradients with respect to the matrix entries are available; eigenvalue and singular-value decompositions have singular gradients at degenerate spectra, which the parametrisations of Section 6 avoid.

Number field and precision

Every algebra is created with two independent switches, `complex_valued` (real or complex matrices) and `precision` ('single' or 'double'), giving the storage types `float32`, `float64`, `complex64` and `complex128`; everything built on the algebra (random operators, states, channels, chains, parametrisations) inherits them. `alg.like(complex_valued=..., precision=...)` creates the same sector structure with other numerics and `op.cast(alg2)` moves an operator there, e.g. to explore in single precision and re-evaluate the interesting cases in double precision.

The field is part of the model. Choosing real matrices is not only a matter of speed: the symmetry group and the random ensembles change. Haar matrices of a real algebra are orthogonal, so moments, level statistics and extremal values refer to $O(n)$ instead of $U(n)$ (e.g. $\mathbb{E}|U_{11}|^4 = 3/(n(n+2))$ instead of $2/(n(n+1))$, Table 4); random real symmetric and complex Hermitian matrices have different level statistics, with mean spacing ratios 0.5321 and 0.6021 for GOE and GUE [17] (computed with the library). Real arithmetic is *exact* (loses nothing) when the problem is real: spectra and eigenvectors of real symmetric matrices, ground states and their entanglement for Hamiltonians without complex hoppings, statistics of time-reversal-invariant systems, and also some extremal problems — for real $X, Y \geq 0$ the supremum of $|\text{Tr}(XUY)|$ over unitaries is attained by a real orthogonal matrix (the polar decomposition of the real matrix YX is real). Complex matrices are *required* for unitary time evolution e^{-iHt} and Lindblad dynamics with a Hamiltonian (the library refuses to run them in a real algebra), for Haar unitaries, CUE/CSE and $SU(n)$, for random pure states and quantities such as Page’s entropy, and for systems without time-reversal symmetry (fluxes, complex couplings).

Precision. Single precision resolves about 10^{-7} relative to the largest entry. It is sufficient for Monte Carlo statistics whose statistical error is much larger (moments, histograms, spacing ratios with 10^3 – 10^5 samples) and for the first, exploratory pass of an optimisation. Double precision is needed when the quantity of interest is small compared with the operators: an inequality checked near equality, small spectral gaps, entropies and logarithms of nearly singular states, contrasts close to 1, comparisons with exact results beyond six digits, long RK4 integrations and optimisation to high relative accuracy. Table 3 shows how drastic the difference can be: the von Neumann entropy of an almost pure state with eigenvalues of order 10^{-9} is wrong by more than 100 % in single precision, and the round-off floor of a trace inequality that holds with equality is 10^{-5} in single and 10^{-13} in double precision.

Cost. An entry takes 4, 8, 8 and 16 bytes for `float32`, `float64`, `complex64` and `complex128`, so memory, and the batch that fits, changes by a factor up to four (`alg.bytes_per_operator(B)` reports it). A complex multiply–add costs about four real ones, and complex eigen- and singular-value solvers are two to four times slower than real ones. Double precision costs about a factor two on CPUs, but on GPUs it depends on the hardware: data-centre GPUs execute FP64 at half the FP32 rate, consumer GPUs at $1/32$ – $1/64$ of it, so that double precision there can be an order of magnitude slower. Table 2 gives the measured ratios for the current run; the recommendation that follows is to use the smallest type that is exact for the model (real if the problem is real) and the precision that the size of the effect requires.

Safeguards. Batched computations make it easy to request hours of work or more memory than the device has. The `cost` module therefore (a) times the first few items of every large batched eigen-, singular-value, QR or exponential decomposition and issues a warning with the expected

type	B/entry	MB/batch	generate	eigh f(A)	SVD A	channel	expm	opt. step
float32	4	1	1	1	1	1	1	1
float64	8	2	1.5	1.5	1.1	1.7	1.7	1.2
complex64	8	2	1.7	1.4	1.5	1.9	2	1.2
complex128	16	4	2.2	1.5	1.7	2.4	2	2.2

Table 2: Memory and time relative to `float32` for $C = 8$ sectors of size $k = 16$ and a batch of 128 operators (quick run on CPU (4 threads); float32 times per sample: generate: $105\ \mu\text{s}$, eigh f(A): $167\ \mu\text{s}$, SVD |A|: $298\ \mu\text{s}$, channel: $30.7\ \mu\text{s}$, expm: $57.4\ \mu\text{s}$, opt. step: $780\ \mu\text{s}$).

type	$\max \ UU^* - 1\ $	$\ \sqrt{A}^2 - A\ /\ A\ $	rel. error $S(\rho)$, $\rho \approx \text{pure}$	$ z $ at equality
float32	8.3×10^{-7}	1.9×10^{-6}	2.3×10^0	2.3×10^{-5}
float64	1.4×10^{-15}	4.0×10^{-15}	5.1×10^{-9}	5.7×10^{-14}
complex64	8.3×10^{-7}	2.2×10^{-6}	2.0×10^0	2.7×10^{-5}
complex128	1.1×10^{-15}	4.4×10^{-15}	5.0×10^{-9}	5.0×10^{-14}

Table 3: Accuracy of the same computations in the four storage types: Haar unitaries ($n = 64$), SVD square root, entropy of $\rho = (1 - \varepsilon)|0\rangle\langle 0| + \varepsilon 1/n$ with $\varepsilon = 10^{-7}$, and the round-off floor of $z = \text{Tr}(Y|X|Y) - \text{Tr}|YXY|$ for commuting X, Y (exactly 0).

total time when it exceeds a threshold (10s by default), (b) compares the estimated peak memory of large allocations with the free memory of the device, warning above 50 % and raising an `InsufficientMemoryError` above 90 %, and (c) reports the expected duration of time-stepping and optimisation loops after their second step, optionally with a progress bar. All thresholds can be changed, and the checks can be switched off locally.

4 Usage by example

The listings of this section are complete programs from `examples/paper`; the test suite runs them, so they stay in sync with the code. They run on a CPU in seconds and use a GPU when available.

Random operators with a prescribed contrast. Listing 1 builds 10^4 positive operators in $M_2 \oplus M_3 \oplus M_4$ whose spectra have contrast 0.6 and whose eigenvectors are Haar-random in every sector, and evaluates traces and spectral functions.

Listing 1: Random operators (`ex1_random_operators.py`).

```

"""Random operators with a prescribed spectrum in M = M_2 (+) M_3 (+) M_4."""
import torch
from torch_vn_algebra import TypeIAlgebra

dev = 'cuda' if torch.cuda.is_available() else 'cpu'
alg = TypeIAlgebra(n_factors=[2, 3, 4], k_factors=[2, 3, 4], device=dev)
B = 10_000                                     # Monte Carlo samples, processed as one batch

def contrast_sampler(delta):
    """Spectra in [lo, 1] with lambda_max = 1, lambda_min = lo, i.e. Michelson contrast delta."""
    lo = (1 - delta) / (1 + delta)

    def sampler(dim):

```

```

        lam = lo + (1 - lo) * torch.rand(B, dim, device=dev)
        lam[:, 0], lam[:, -1] = 1.0, lo
        return lam
    return sampler

# A_c = U_c diag(lambda) U_c^*, with independent Haar unitaries U_c in every block and sample
X = alg.operator_from_eigenvalues(contrast_sampler(0.6), batch_size=B,
                                force_positive=True, force_self_adjoint=True)
print(X.matrix.shape) # (10000, 3, 4, 4): batch, sector, padded block
print(X.michelson_contrast[:4]) # 0.6 for every sample
print(X.Tr_blunt().real.mean(), X.Tr_norm().mean(), X.tau_vN().mean())

# functional calculus through the spectral theorem, block by block
R = X.sqrt()
print((R @ R - X).frobenius_norm().max()) # ~1e-6 in single precision
print(X.apply_function(torch.log).tau_vN().mean()) # tau_vN(log X)

```

States, measurement and trace conventions. Listing 2 conditions random states on the outcome of a projective measurement (Lüders rule), which moves all probability into one sector, and checks that expectation values do not depend on the trace used to represent the state.

Listing 2: States and conditioning (ex2_states_measurement.py).

```

"""States on M, sector probabilities, Lueders conditioning and trace conventions."""
import torch
from torch_vn_algebra import TypeIAlgebra, DensityMatrix, states

dev = 'cuda' if torch.cuda.is_available() else 'cpu'
alg = TypeIAlgebra([2, 3], [2, 3], device=dev) # two superselection sectors

rho = states.random_density_matrix(alg, batch_size=4) # Hilbert-Schmidt random states
print(rho.sector_probabilities()) # p_c = Tr rho_c, rows sum to 1

# measure the projection onto the first basis vector of sector 0
P = alg.from_blocks([torch.diag(torch.tensor([1.0, 0.0])), torch.zeros(3, 3)])
post, prob = rho.condition_on(P) # rho|P = P rho P / Tr(rho P)
print(prob, post.sector_probabilities()) # all weight now in sector 0

# the same state as a density w.r.t. tau_vN: omega(A) = tau_vN(d A) = Tr(rho A)
A = alg.operator_from_eigenvalues(lambda d: torch.randn(4, d), batch_size=4,
                                force_self_adjoint=True)

d_tau = rho.density('tau_vN')
print(rho.expectation(A).real, (alg.operator(d_tau) @ A).tau_vN().real)
back = DensityMatrix.from_density(alg, d_tau, trace='tau_vN')
print((back.matrix - rho.matrix).abs().max())

```

A map between sectors. Two fermionic modes have sectors $N = 0, 1, 2$ of dimensions 1, 2, 1. Particle loss with Kraus operators $\sqrt{1-g}N$, $\sqrt{g}a_1$, $\sqrt{g}a_2$ maps sector N to N and $N-1$. Listing 3 builds it from its blocks; the classical part is the transition matrix of a death process,

$$T = \begin{pmatrix} 1 & g & 0 \\ 0 & 1-g & 2g \\ 0 & 0 & 1-2g \end{pmatrix},$$

which the program prints. The last lines illustrate a simple fact: a unital positive map cannot increase the Michelson contrast, since $\lambda_{\min}1 \leq A \leq \lambda_{\max}1$ implies the same bounds for $\Phi(A)$; the non-unital loss channel increases it for most samples.

Listing 3: Particle loss between sectors (`ex3_channels.py`).

```
"""Particle loss between charge sectors of two fermionic modes, as an InterSectorChannel."""
import torch
from torch_vn_algebra import TypeIAlgebra, InterSectorChannel, channels, states

dev = 'cuda' if torch.cuda.is_available() else 'cpu'
# sectors N = 0, 1, 2 with bases |00>, |10>, |01>, |11>
alg = TypeIAlgebra([1, 2, 1], [1, 2, 1], device=dev)
g = 0.2 # loss probability per particle
s, z = g ** 0.5, torch.zeros
# Kraus operators K_0 = sqrt(1 - g N), K_1 = sqrt(g) a_1, K_2 = sqrt(g) a_2, split into blocks (d, c)
blocks = {
    (0, 0): [torch.ones(1, 1)],
    (1, 1): [(1 - g) ** 0.5 * torch.eye(2)],
    (2, 2): [(1 - 2 * g) ** 0.5 * torch.ones(1, 1)],
    (0, 1): [z(1, 2), s * torch.tensor([[1.0, 0.0]]), s * torch.tensor([[0.0, 1.0]])],
    (1, 2): [z(2, 1), s * torch.tensor([[0.0], [1.0]]), s * torch.tensor([[1.0], [0.0]])],
}
loss = InterSectorChannel.from_blocks(alg, alg, blocks)
print(loss.is_trace_preserving())
print(loss.transition_matrix()[0]) # classical death process on N

rho = states.random_density_matrix(alg, batch_size=3)
print(loss(rho).sector_probabilities()) # still a DensityMatrix

# unital channels cannot increase the Michelson contrast, non-unital ones can
X = alg.operator_from_eigenvalues(lambda d: 0.5 + torch.rand(1000, d, device=dev),
                                batch_size=1000, force_positive=True)
mix = channels.random_mixed_unitary_channel(alg, 3, batch_size=1000)
print(bool((mix(X).michelson_contrast <= X.michelson_contrast + 1e-6).all()))
print((loss(X).michelson_contrast > X.michelson_contrast).float().mean())
```

Open-system dynamics. Listing 4 integrates spontaneous emission of a two-level atom with the RK4 solver and with the exact exponential of the Lindblad generator, and compares both with $P_e(t) = P_e(0)e^{-\gamma t}$ and $|\rho_{eg}(t)| = |\rho_{eg}(0)|e^{-\gamma t/2}$.

Listing 4: Spontaneous emission (`ex4_dynamics.py`).

```
"""Spontaneous emission of a two-level atom: Lindblad evolution against the exact solution."""
import math
import torch
from torch_vn_algebra import TypeIAlgebra, DensityMatrix, dynamics

dev = 'cuda' if torch.cuda.is_available() else 'cpu'
alg = TypeIAlgebra([2], [2], device=dev, precision='double')
omega, gamma = 1.0, 0.3
H = alg.from_blocks([torch.diag(torch.tensor([0.5 * omega, -0.5 * omega]))]) # basis |e>, |g>
L = alg.from_blocks([torch.tensor([[0.0, 0.0], [1.0, 0.0]])]) # sigma_- = |g><e|
psi = torch.tensor([0.6, 0.8], dtype=torch.complex128)
rho0 = DensityMatrix(alg, matrix=torch.outer(psi, psi.conj())[None, None].to(dev))

times = [0.0, 1.0, 2.0, 5.0]
rk4 = dynamics.lindblad_evolve(rho0, H, [L], times, rates=[gamma], substeps=100)
for t, rho in zip(times, rk4):
```

```

exact = dynamics.lindblad_channel(alg, H, [L], t=t, rates=[gamma])(rho0) # exp(tL)
p_e = 0.36 * math.exp(-gamma * t) # analytic excited population
coh = 0.48 * math.exp(-gamma * t / 2) # analytic |rho_eg|
m = rho.matrix[0, 0]
print(f"t={t}: P_e {m[0, 0].real:.6f} (exact {p_e:.6f}), |rho_eg| {m[0, 1].abs():.6f} "
      f"(exact {coh:.6f}), RK4 vs exp(tL): {(rho.matrix - exact.matrix).abs().max():.1e}")

```

Tensor products. Listing 5 estimates the mean entanglement entropy of random pure states on $\mathbb{C}^m \otimes \mathbb{C}^n$ and compares it with Page’s formula [13].

Listing 5: Page entropy (ex5_page_entropy.py).

```

"""Average entanglement entropy of random pure states on C^m (x) C^n (Page, 1993)."""
import torch
from torch_vn_algebra import TypeIAlgebra, tensor_product, partial_trace, states

dev = 'cuda' if torch.cuda.is_available() else 'cpu'
m, n, B = 4, 8, 20_000
A = TypeIAlgebra([m], [m], device=dev, precision='double')
Bsys = TypeIAlgebra([n], [n], device=dev, precision='double')
AB = tensor_product(A, Bsys)

psi = states.random_density_matrix(AB, batch_size=B, rank=1) # Haar-random pure states
S = states.von_neumann_entropy(partial_trace(psi, keep=1)) # entropy of rho_A

page = sum(1.0 / k for k in range(n + 1, m * n + 1)) - (m - 1) / (2 * n)
print(f"<S_A> = {S.mean():.5f} +- {S.std() / B ** 0.5:.5f}, Page: {page:.5f}")

```

Searching for a violation. Listing 6 compares Haar sampling with batched gradient ascent for $\sup_U (|\text{Tr}(XUY)| - \text{Tr}(XY))$, whose exact value $\|YX\|_1 - \text{Tr}(XY)$ is positive for non-commuting $X, Y \geq 0$. Section 6 studies this systematically.

Listing 6: Sampling versus optimisation (ex6_inequality_search.py).

```

"""Searching for violations of |Tr(XUY)| <= Tr(XY): random sampling versus gradient ascent."""
import torch
from torch_vn_algebra import TypeIAlgebra

dev = 'cuda' if torch.cuda.is_available() else 'cpu'
torch.manual_seed(0)
k = 16
alg = TypeIAlgebra([k], [k], device=dev, precision='double')
pos = lambda d: 0.2 + torch.rand(1, d)
X = alg.operator_from_eigenvalues(pos, force_positive=True, force_self_adjoint=True)
Y = alg.operator_from_eigenvalues(pos, force_positive=True, force_self_adjoint=True)
trXY = (X @ Y).trace.real

def z(U):
    # z(U) = |Tr(XUY)| - Tr(XY), batched over U
    return (X @ U @ Y).trace.abs() - trXY

# 1) Monte Carlo over Haar unitaries
print("sampling, 10^5 Haar U:", z(alg.random_unitary_operator(100_000)).max().item())

# 2) gradient ascent over U = U0 exp(G - G^*), 64 random starts optimised in parallel;
# the exponential chart is re-centred at the current point every 50 steps

```

```

U0 = alg.random_unitary_operator(64).matrix
for _ in range(10):
    G = torch.zeros(64, 1, k, k, dtype=alg.hilbert.dtype, device=dev, requires_grad=True)
    opt = torch.optim.Adam([G], lr=0.05)
    for step in range(50):
        U = alg.operator(U0) @ alg.operator(G - G.conj().transpose(-2, -1)).expm()
        loss = -z(U).sum()
        opt.zero_grad()
        loss.backward()
        opt.step()
    U0 = U.matrix.detach()
print("gradient ascent:      ", z(alg.operator(U0)).max().item())

# exact supremum: sup_U |Tr(U YX)| = ||YX||_1
print("exact sup:          ", ((Y @ X).trace_norm() - trXY).item())

```

Spin chains. Listing 7 works with a Heisenberg ring of ten spins. A uniform field moves the ground state through the sectors until the chain is fully polarised above the saturation field $h_s = 2J$; 100 realisations of random fields are diagonalised in one batch to compute the mean level-spacing ratio; and the entanglement entropy of the ground state is computed from reduced states. Section 7 checks these tools against exact results.

Listing 7: Spin chains (ex7_spin_chain.py).

```

"""Spin chains with conserved S^z: sectors, disorder averages and entanglement."""
import torch
from torch_vn_algebra import SpinChain, DensityMatrix

dev = 'cuda' if torch.cuda.is_available() else 'cpu'
chain = SpinChain(L=10, boundary='periodic', device=dev)          # sectors N = 0..10, dims binom(10, N)
print(chain.algebra.k_factors)

# Heisenberg ring in a uniform field: the ground state changes sector as h grows
for h in (0.0, 1.0, 2.5):
    w = [blk[0] for blk, _ in chain.heisenberg(h=h).eigh()]
    print(f"h={h}: ground state in sector N={min(range(11), key=lambda N: w[N].min())}")

# random fields h_i ~ U[-W, W]: 100 disorder realisations in one batch, sector N = 5 only
for W in (0.5, 8.0):
    H = chain.random_field_heisenberg(W, batch_size=100, sector=5)
    E, _ = H.eigh()[0]                                           # (100, 252)
    r = SpinChain.level_spacing_ratio(E[:, 63:189]).mean()
    print(f"W={W}: <r> = {r:.3f}    (GOE 0.531, Poisson 0.386)")

# entanglement entropy of the ground state of the clean chain, left block of n sites
E, V = chain.heisenberg(sector=5).eigh()[0]
psi = V[0, :, 0]
rho = DensityMatrix(chain.sector_algebra(5), matrix=torch.outer(psi, psi.conj())[None, None])
print([round(chain.entanglement_entropy(rho, n, sector=5).item(), 4) for n in range(1, 6)])

```

5 Validation against exact results

Table 4 compares Monte Carlo estimates produced with the library (double precision) with exact values: fourth moments of Haar unitaries and orthogonal matrices; the Weingarten-calculus value

$\mathbb{E}\tau((AUBU^*)^2) = -1/(N^2 - 1)$ for traceless involutions $A = B$ and Haar U ; Page's formula for $\langle S(\rho_A) \rangle$; and the mean purity $(k + r)/(kr + 1)$ of induced random states of rank r [14]. The last column is the deviation in units of the standard error; all are of order one. For the circular ensembles the Wigner surmise is only an approximation to the exact spacing distribution, so no z -score is given; Figure 1 shows the distributions, with L^1 distances 0.031, 0.021 and 0.029 for $\beta = 1, 2, 4$ from the surmise; level repulsion $P(s) \sim s^\beta$ is clearly visible. The spontaneous-emission example agrees with the analytic solution to 2.2×10^{-16} (exact exponential) and 2.1×10^{-8} (RK4).

quantity	parameters	exact	Monte Carlo	std. error	z
$\mathbb{E} U_{11} ^4, U(n)$	$n = 2$	0.333333	0.335015	1.5×10^{-3}	+1.1
$\mathbb{E} U_{11} ^2, U(n)$	$n = 2$	0.5	0.5014	1.4×10^{-3}	+1.0
$\mathbb{E} U_{11} ^4, U(n)$	$n = 4$	0.1	0.0999686	6.8×10^{-4}	-0.0
$\mathbb{E} U_{11} ^4, U(n)$	$n = 8$	0.0277778	0.0277718	2.4×10^{-4}	-0.0
$\mathbb{E} U_{11} ^4, U(n)$	$n = 16$	0.00735294	0.00737778	7.1×10^{-5}	+0.4
$\mathbb{E} U_{11} ^4, U(n)$	$n = 32$	0.00189394	0.00188255	2.0×10^{-5}	-0.6
$\mathbb{E} U_{11} ^2, U(n)$	$n = 32$	0.03125	0.031172	1.5×10^{-4}	-0.5
$\mathbb{E} U_{11} ^4, O(n)$	$n = 2$	0.375	0.373641	1.8×10^{-3}	-0.7
$\mathbb{E} U_{11} ^2, O(n)$	$n = 2$	0.5	0.498956	1.8×10^{-3}	-0.6
$\mathbb{E} U_{11} ^4, O(n)$	$n = 4$	0.125	0.12365	9.8×10^{-4}	-1.4
$\mathbb{E} U_{11} ^4, O(n)$	$n = 8$	0.0375	0.0378452	4.0×10^{-4}	+0.9
$\mathbb{E} U_{11} ^4, O(n)$	$n = 16$	0.0104167	0.0104973	1.4×10^{-4}	+0.6
$\mathbb{E} U_{11} ^4, O(n)$	$n = 32$	0.00275735	0.0027189	3.9×10^{-5}	-1.0
$\mathbb{E} U_{11} ^2, O(n)$	$n = 32$	0.03125	0.0311461	2.1×10^{-4}	-0.5
$\mathbb{E}\tau((AUBU^*)^2)$	$N = 4$	-0.0666667	-0.0668132	1.8×10^{-3}	-0.1
$\mathbb{E}\tau((AUBU^*)^2)$	$N = 16$	-0.00392157	-0.00342506	4.0×10^{-4}	+1.3
$\mathbb{E}\tau((AUBU^*)^2)$	$N = 64$	-0.0002442	-0.000192229	2.0×10^{-4}	+0.3
$\langle s^2 \rangle$, COE	$N = 32$	1.27324	1.28476	5.2×10^{-3}	—
$\langle s^2 \rangle$, CUE	$N = 32$	1.1781	1.17783	3.8×10^{-3}	—
$\langle s^2 \rangle$, CSE	$N = 32$	1.10447	1.10321	3.9×10^{-3}	—
$\langle S(\rho_A) \rangle$	$m = 2, n = 8$	0.600372	0.600235	9.6×10^{-4}	-0.1
$\langle S(\rho_A) \rangle$	$m = 4, n = 8$	1.15314	1.15389	1.0×10^{-3}	+0.7
$\langle S(\rho_A) \rangle$	$m = 8, n = 8$	1.58853	1.58838	8.7×10^{-4}	-0.2
$\mathbb{E}\text{Tr}\rho^2$	$k = 4, r = 2$	0.666667	0.666228	1.0×10^{-3}	-0.4
$\mathbb{E}\text{Tr}\rho^2$	$k = 4, r = 4$	0.470588	0.470432	6.8×10^{-4}	-0.2
$\mathbb{E}\text{Tr}\rho^2$	$k = 16, r = 4$	0.307692	0.307736	2.0×10^{-4}	+0.2
$\mathbb{E}\text{Tr}\rho^2$	$k = 16, r = 16$	0.124514	0.124468	5.4×10^{-5}	-0.8

Table 4: Monte Carlo estimates versus exact values (quick run on CPU (4 threads), PyTorch 2.14.1+cu130).

6 From sampling to optimisation

The first version of this work [1] studied three trace functionals by Monte Carlo:

$$\begin{aligned}
\text{Exp. 1: } z &= |\text{Tr}(XUY)| - \text{Tr}(XY), & X, Y \geq 0, U \in \mathcal{M} \text{ unitary,} \\
\text{Exp. 2: } z &= \text{Tr}|XY| - \text{Tr}(|X||Y|), & X = UX_0, Y = VY_0, X_0, Y_0 \geq 0, \\
\text{Exp. 3: } z &= \text{Tr}(Y|X|Y) - \text{Tr}|YXY|, & X = X^*, Y \geq 0,
\end{aligned}$$

with $\text{Tr} = \text{Tr}_{\text{blunt}}$, operators normalised to $\lambda_{\max} = 1$, and the contrasts of the operators as parameters. Two observations motivate a different approach.

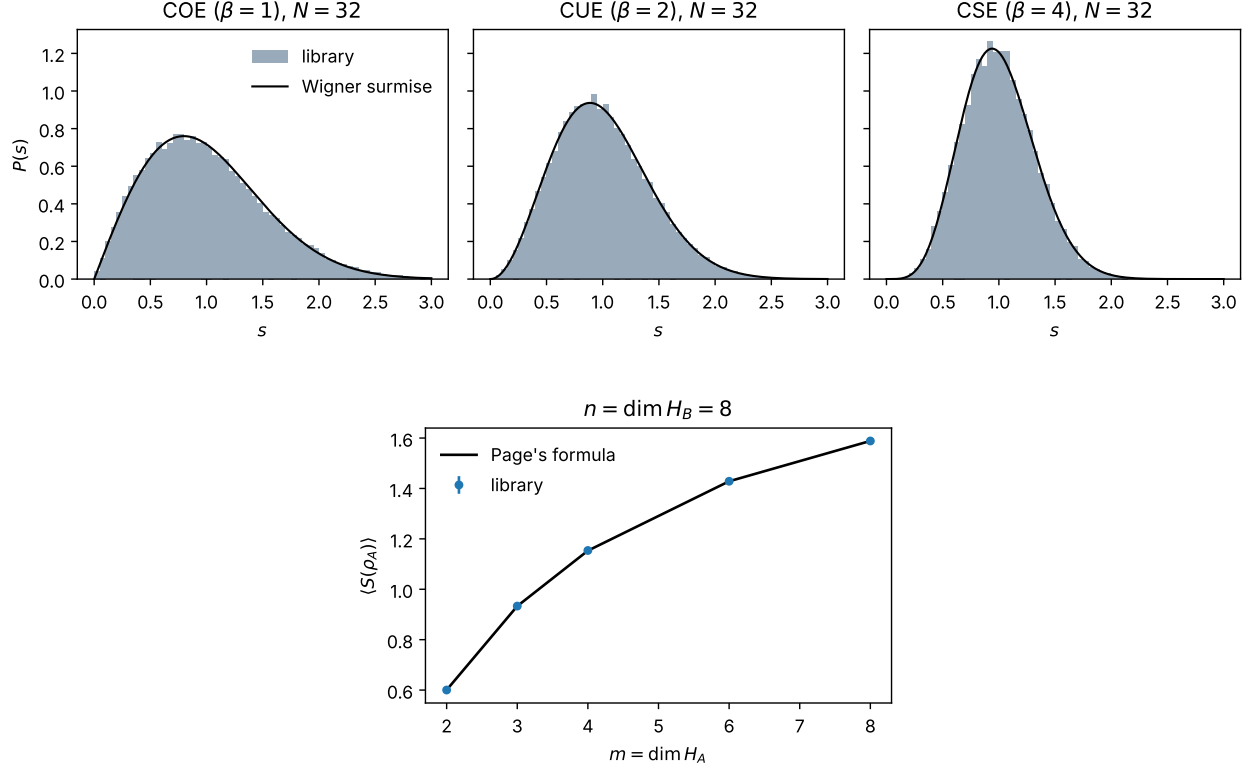


Figure 1: Top: nearest-neighbour spacings of COE, CUE and CSE eigenphases generated by the library, against the Wigner surmise. Bottom: mean entanglement entropy of random pure states against Page’s formula (error bars: two standard errors).

Sampling misses the extremes. For fixed X, Y the supremum in Exp. 1 is known exactly, $\sup_U z = \|YX\|_1 - \text{Tr}(XY) > 0$. Table 5 and Figure 2 show that the best of 20 000 Haar samples is already negative for $k \geq 4$ and moves further away from the supremum as k grows, while batched gradient ascent (32 starts, 500 steps) finds it to a relative accuracy of 10^{-5} or better. This is concentration of measure: the unitaries close to the maximiser form a set whose Haar measure decays rapidly with the dimension. In particular, the absence of violations in Monte Carlo samples for $k = 16$ reported in [1] is a property of the sampling, not of the inequality.

k	violation found		$\sup z / \text{Tr } XY$	best sample $z / \text{Tr } XY$	ascent rel. error		time [s]	
	sampling	ascent			median	max	sampl.	ascent
2	2/4	4/4	4.4×10^{-5}	-0.00	5.4×10^{-8}	1.7×10^{-5}	0.015	0.86
4	0/4	4/4	2.0×10^{-3}	-0.19	9.4×10^{-8}	7.5×10^{-7}	0.049	0.86
8	0/4	4/4	3.2×10^{-3}	-0.58	5.0×10^{-7}	2.1×10^{-6}	0.18	1.2
16	0/4	4/4	4.5×10^{-3}	-0.78	2.6×10^{-6}	4.4×10^{-6}	0.77	2.6

Table 5: Exp. 1 for fixed random X, Y : Haar sampling versus gradient ascent (medians over 4 pairs; quick run on CPU (4 threads)).

Constrained optimisation. The natural objects are the envelopes

$$\bar{z}(\delta) = \sup\{z : \Delta(X) = \delta\}, \quad \underline{z}(\delta) = \inf\{z : \Delta(X) = \delta\},$$

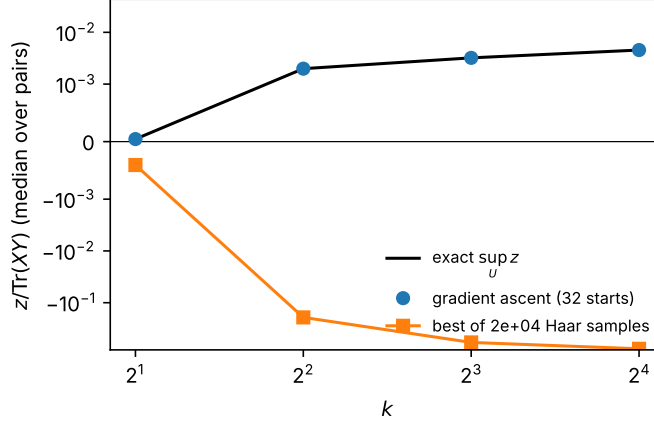


Figure 2: Best z found by sampling and by gradient ascent, relative to $\text{Tr}(XY)$, against the exact supremum (symmetric log scale).

(and the same with $\Delta(Y)$, or with both contrasts prescribed), the supremum being taken over all remaining variables. The module `optimize` parametrises the constraint set exactly: the spectrum is $\lambda_{\max} = 1$, $\lambda_{\min} = (1 - \delta)/(1 + \delta)$ and the other eigenvalues in between (through a sigmoid), the eigenvectors are $U = U_0 e^{G - G^*}$ with a Haar-random U_0 per start and a chart that is re-centred after every round, and for Exp. 3 the signs of the eigenvalues of X are drawn per start and kept fixed. Every grid point and every start is one element of the batch, so the whole envelope is computed in a single batched optimisation.

Figures 3 and 4 and Table 6 show the result. Every Monte Carlo sample of [1] lies between the computed envelopes; samples that fall outside the linear interpolation between grid points are re-optimised at their exact contrast, after which none remains outside. The envelopes are much wider than the sample clouds, already for $k = 2$.

Two envelopes can be checked analytically. In Exp. 3, $z \geq 0$ always:

Proposition 1. *For $X = X^*$ and $Y \geq 0$, $\text{Tr} |YXY| \leq \text{Tr}(Y|X|Y)$, with equality if Y commutes with X .*

Proof. With $X = X_+ - X_-$, $|X| = X_+ + X_-$, the triangle inequality for the trace norm gives $\|YX_+Y - YX_-Y\|_1 \leq \text{Tr}(YX_+Y) + \text{Tr}(YX_-Y)$. If $XY = YX$, then $|YXY| = Y^2|X|$. $\square$

The computed lower envelope of Exp. 3 is zero to machine precision. In Exp. 1, the lower envelope for prescribed $\Delta(X) = \delta$ equals $-(kC - 2\delta/(1 + \delta))$ when all sectors have size $k \geq 2$: $\text{Tr}(XY) \leq \text{Tr} X \leq kC - 1 + \lambda_{\min}(X)$ with equality for $Y = 1$, while $|\text{Tr}(UYX)|$ can be made zero by a cyclic permutation of the eigenbasis of X_c in every sector. The optimiser reproduces this curve to a relative deviation of $8.1\text{e-}03$. The upper envelopes of all three experiments are, to our knowledge, not known in closed form; the computed curves are lower bounds for them (a local search) and suggest conjectures for sharp contrast-dependent inequalities, which we leave for future work.

7 Symmetric tensor products and spin chains

The basis of `SpinChain` is, by construction, that of the iterated fused product of single sites: the tests check that S_i^z built by the chain coincides with $S_i^z \otimes 1$ formed with `kron` in the fused product of the

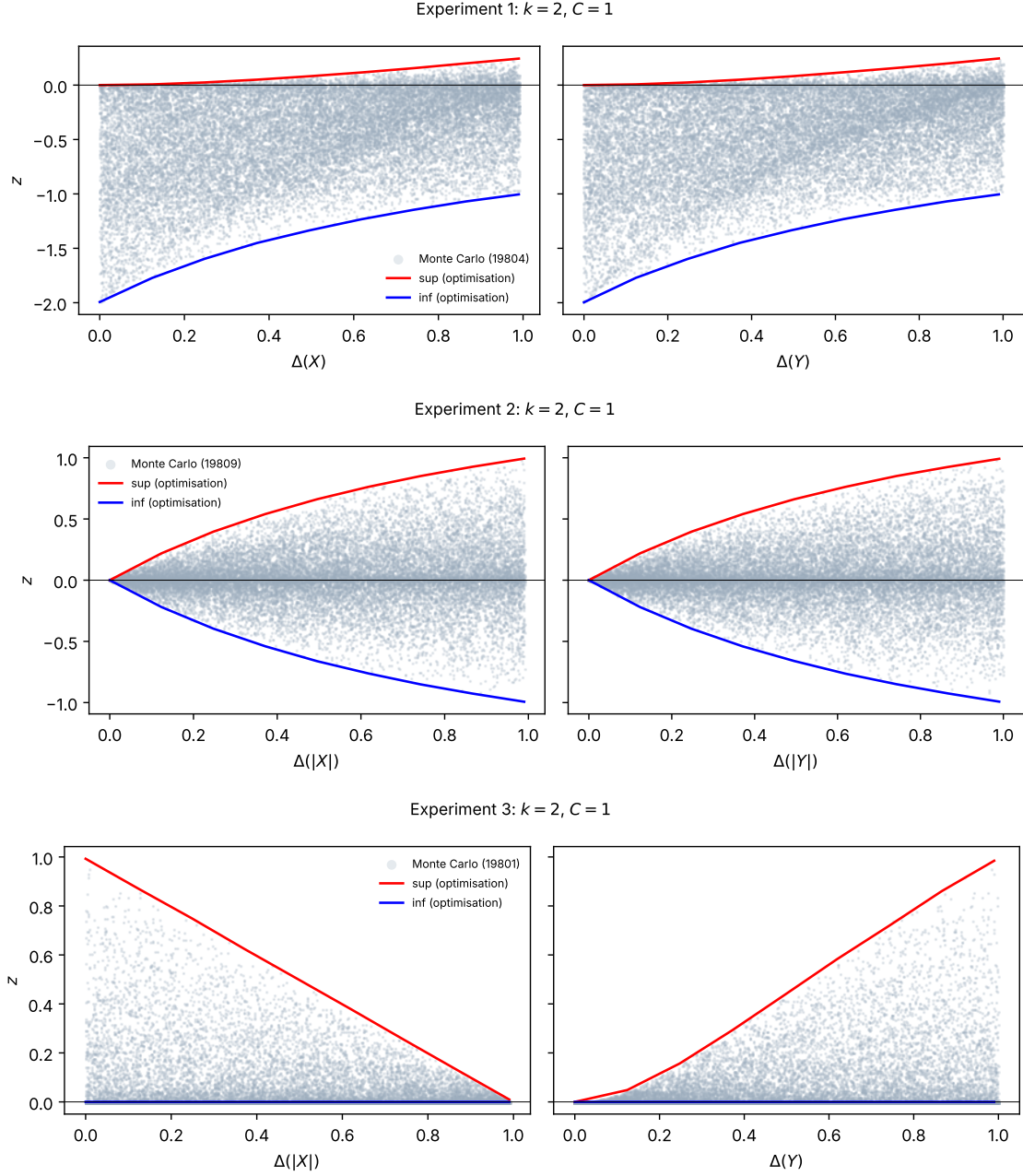


Figure 3: Monte Carlo samples (grey) and the upper (red) and lower (blue) envelopes of z as functions of one contrast, the other one being optimised as well.

Exp.	k	C	inf z	sup z	Monte Carlo range of z	outside
1	2	1	-1.994	+0.2436	$[-1.98, +0.225]$	0
1	2	2	-3.991	+0.4765	$[-3.39, +0.328]$	0
1	4	1	-3.994	+0.472	—	—
2	2	1	-0.9931	+0.9934	$[-0.921, +0.975]$	0
2	2	2	-1.956	+1.963	$[-1.43, +1.72]$	0
2	4	1	-1.961	+1.957	—	—
3	2	1	0	+0.9927	$[0, +0.946]$	0
3	2	2	0	+1.968	$[0, +1.39]$	0
3	4	1	0	+1.962	—	—

Table 6: Global extremes of z over all contrasts found by constrained optimisation, the range of the Monte Carlo samples, and the number of samples outside the envelopes after re-optimisation (12 starts per grid point, 240 steps; quick run on CPU (4 threads)).

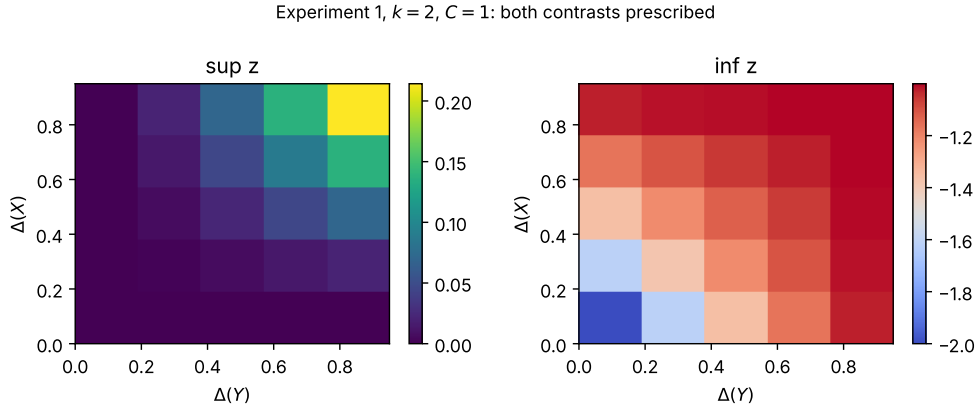


Figure 4: Exp. 1, $k = 2$, $C = 1$: sup z and inf z with both contrasts prescribed.

algebra of **SpinChain(L-1)** with a single site. Beyond that, integrable chains provide exact results for every part of the module (Table 7, Figure 5; script `scripts/paper/chains.py`):

- *Free fermions.* By the Jordan–Wigner transformation the open XX chain ($\Delta = 0$) is a free-fermion model with single-particle energies $\varepsilon_k = J \cos \frac{\pi k}{L+1}$, so the spectrum of sector N consists of all sums of N distinct ε_k . For $L = 10$ all 2^L levels agree to 8.0×10^{-15} .
- *Heisenberg ring.* Ground-state energies agree with the tabulated exact values to the nine digits of the table; E_0/L approaches the Bethe-ansatz value $\frac{1}{4} - \ln 2 \approx -0.4431$. Every level of sector $N < L/2$ reappears in sector $N + 1$ (SU(2) multiplets), up to 1.5×10^{-14} .
- *Dynamics.* The melting of a domain wall $|\uparrow \cdots \uparrow \downarrow \cdots \downarrow\rangle$ in the XX chain, evolved in the sector $N = L/2$ (dimension 924 for $L = 12$), reproduces the single-particle prediction $\langle n_i(t) \rangle = [e^{-iht}C(0)e^{iht}]_{ii}$ to 5.0×10^{-15} .
- *Entanglement.* The entropy of a block of n sites of the XX ground state, obtained with `reduced_state`, agrees with Peschel’s formula $S = -\sum_\nu [\nu \ln \nu + (1 - \nu) \ln(1 - \nu)]$ over the eigenvalues ν of the restricted correlation matrix [15], including the parity oscillations of open chains, to 3.3×10^{-12} .

- *Disorder.* For the Heisenberg chain with random fields $h_i \in [-W, W]$ the mean ratio of consecutive level spacings in the middle of the spectrum crosses over from the GOE value 0.5307 to the Poisson value $2 \ln 2 - 1 \approx 0.3863$ [16, 17] as W grows, the hallmark of many-body localisation. Here the batch dimension holds the disorder realisations (40 per point).

L	$\dim (N = L/2)$	E_0 (library)	E_0 (exact)	difference	E_0/L
4	6	-2.000000000	-2.000000000	1.0×10^{-16}	-0.50000
6	20	-2.802775638	-2.802775638	2.7×10^{-10}	-0.46713
8	70	-3.651093409	-3.651093409	6.3×10^{-11}	-0.45639
10	252	-4.515446354	-4.515446354	4.9×10^{-10}	-0.45154
12	924	-5.387390917	-5.387390917	4.5×10^{-10}	-0.44895

Table 7: Ground-state energy of the Heisenberg ring $H = \sum_i \mathbf{S}_i \cdot \mathbf{S}_{i+1}$; exact values to nine digits (quick run on CPU (4 threads)).

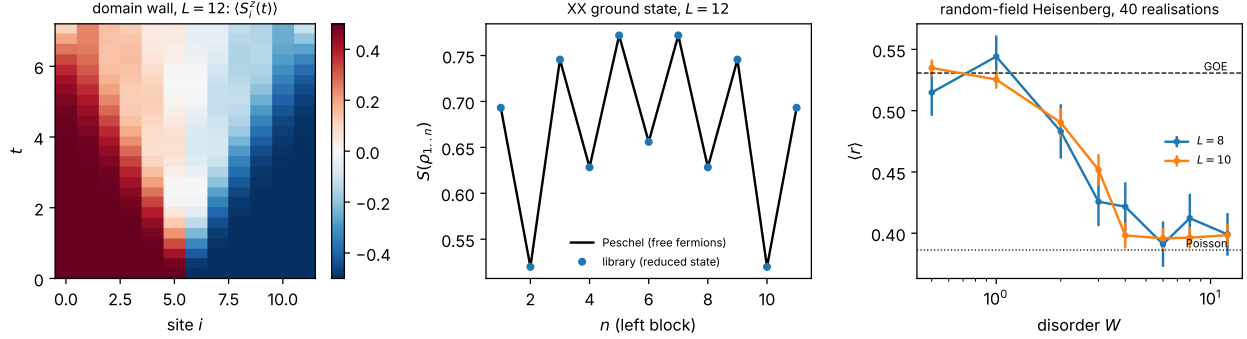


Figure 5: Left: $\langle S_i^z(t) \rangle$ after a domain-wall quench of the XX chain (light cone). Middle: entanglement entropy of the XX ground state, library vs. Peschel’s formula. Right: mean level-spacing ratio of the random-field Heisenberg chain, with the GOE and Poisson values.

8 Performance

Table 8 and Figure 6 report the wall-clock time per sample of library operations (not of isolated kernels), in single precision, with the batch size chosen so that the batch holds about 2^{22} matrix entries (quick run on CPU (4 threads); CPU with 4 threads). Expected behaviour: for single small blocks the CPU is faster, because kernel launches dominate; the GPU gains for many sectors and for the matrix products in channels and in the optimisation step; batched symmetric eigen- and singular-value solvers lose much of their advantage from $k \approx 64$ on, where the GPU libraries switch from batched small-matrix kernels to general ones.

9 Limitations and roadmap

- *Layout.* Blocks are padded to $k_{\max}$. For very different sector sizes (binomial sectors of spin chains) one works in single sectors; operators acting on all sectors at once would need a bucketed layout, which is not implemented yet.

operation	k	C	batch	time per sample [μ s]
generate	4	1	4096	1.33
f(A) eigh	4	1	4096	3.19
A SVD	4	1	4096	6.14
contrast	4	1	4096	2.07
channel	4	1	4096	0.595
exp(tL)	4	1	4096	59.6
opt. step	4	1	4096	13.6
generate	4	16	1024	28.1
f(A) eigh	4	16	1024	44.1
A SVD	4	16	1024	90
contrast	4	16	1024	24.3
channel	4	16	1024	7.81
exp(tL)	4	16	1024	1.42e+03
opt. step	4	16	1024	314
generate	16	1	1024	18.1
f(A) eigh	16	1	1024	35.8
A SVD	16	1	1024	67.2
contrast	16	1	1024	14
channel	16	1	1024	2.96
exp(tL)	16	1	1024	1.67e+04
opt. step	16	1	1024	205
generate	16	16	64	439
f(A) eigh	16	16	64	495
A SVD	16	16	64	938
contrast	16	16	64	200
channel	16	16	64	46.7
exp(tL)	16	16	64	2.48e+05
opt. step	16	16	64	3e+03

Table 8: Time per sample of library operations.

- *Symmetries.* Only abelian charges with an arbitrary fusion rule are supported; non-abelian symmetries (SU(2) multiplets, Clebsch–Gordan structure) and lattice translations are not used to reduce blocks further. Chains are treated by dense diagonalisation, i.e. up to about 16–18 spins.
- *Optimisation.* The search is local; envelopes are lower bounds for suprema (upper bounds for infima). Gradients of spectral functions are singular at degenerate spectra; the parametrisations avoid this for the contrast constraint, but not for every user-defined objective.
- *Scale.* Choi matrices, superoperators and $e^{t\mathcal{L}}$ work with $k_c^2 \times k_c^2$ matrices and are limited to blocks of a few tens; RK4 integration scales to larger blocks.
- *Scope.* Finite dimensions and Type I only; no tensor-network or sparse methods.

10 Availability and reproducibility

The code (MIT licence) is available at https://github.com/andre-a-no/von_neumann_type_i. `bash scripts/paper/run_all.sh full cuda` regenerates every table and figure of this text on a GPU; `bash scripts/paper/run_all.sh` runs a reduced version on a CPU in a few minutes. The tables state on which device and in which mode they were produced.

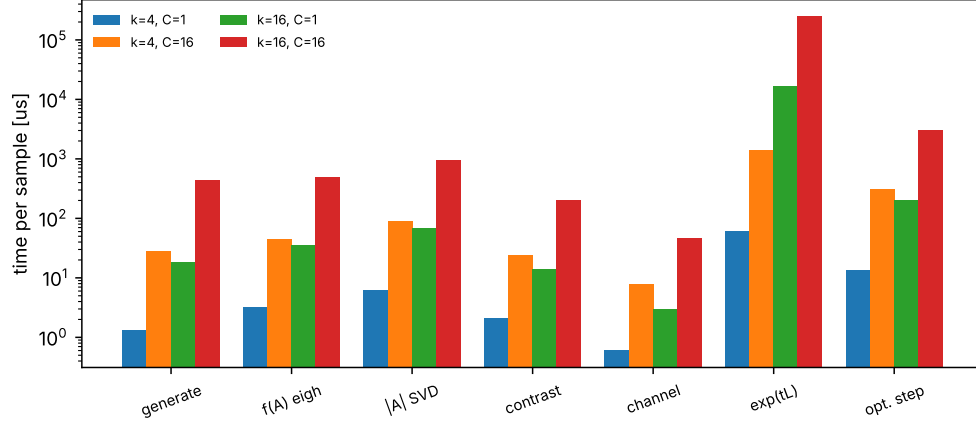


Figure 6: Time per sample of library operations.

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